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Field-Scale Rice Area and Yield Mapping in Sri Lanka with Optical Remote Sensing and Limited Training Data

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Abstract

Rice is a staple crop for over half the world's population, and accurate, timely information on its planted area and production is crucial for food security and agricultural policy, particularly in developing nations like Sri Lanka. However, reliable rice monitoring in regions like Sri Lanka faces significant challenges due to frequent cloud cover and the fragmented nature of smallholder farms. This research introduces a novel, cost-effective method for mapping rice-planted area and yield at field scales in Sri Lanka using optical satellite data. The rice-planted fields were identified and mapped using a phenologically tuned image classification algorithm that highlights rice presence by observing water occurrence during transplanting and vegetation activity during subsequent crop growth. To estimate yields, a random forest regression model was trained at the district level by incorporating a satellite-derived chlorophyll index and environmental variables and subsequently applied at the field level. The approach has enabled the creation of two decades (2000–2022) of reliable, field-scale rice area and yield estimates, achieving map accuracies between 70% and over 90% and yield estimates with less than 20% error. These highly granular results, which are not available through traditional surveys, show a strong correlation with government statistics. They also demonstrate the advantages of a rule-based, phenology-driven classification over purely statistical machine learning models for long-term consistency in dynamic agricultural environments. This work highlights the significant potential of remote sensing to provide accurate and detailed insights into rice cultivation, supporting policy decisions and enhancing food security in Sri Lanka and other cloud-prone regions.

Keywords: rice; Sri Lanka; Landsat; Sentinel-2; irrigation; phenology

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1. Introduction

Rice (*Oryza sativa* L.) is an important staple food crop, providing the predominant source of daily nourishment for more than 3.5 billion people around the world and making up about 16.5% of global caloric intake [1,2]. The global production of rice is dominated by Asian countries, given their warm and humid climate. Rice (or paddy rice) is an important crop in Sri Lanka, where its importance goes far beyond its status as a primary food source and plays an outsized role in the country's cultural identity, tradition, and politics. Daily per capita rice consumption in Sri Lanka (at around 300 g per day) is reported to be twice that of the global average [3], which illustrates the role of rice in the Sri Lankan diet and economics.

The importance of rice in Sri Lanka is also reflected in carefully designed and successfully executed annual paddy surveys that produce reliable rice area and yield estimates at the district level [4]. These surveys play a vital role in providing official statistics on rice production in Sri Lanka at the national and district levels. However, while these surveys serve their purpose for broader policy and planning, they often fall short in capturing the nuances and complexities at finer scales, particularly for smallholder farmers, and, more importantly, their variation associated with weather, management, and policy. This results in a missed opportunity to quantify phenomena such as agricultural water consumption, the impacts of climate change on rice production, the effects of pests and diseases, the effectiveness of nutrient applications, and methane emissions. To this end, the spatial distribution of paddy rice, particularly that derived from remote sensing, constitutes valuable information to address various agricultural management, environmental impact, water use, and food security questions. Moreover, compared to ground surveys, remote sensing provides a cost-effective and reliable alternative to capture rice-planted areas. Our work presented here describes a remote sensing-based methodology to map rice-growing areas and their yields at field scales using optical observations. Remote sensing has been used in several studies to map rice-planted areas [5–10]. Successful rice identification and mapping studies rely on unique remotely sensed reflectance or Synthetic Aperture Radar (SAR) backscatter temporal profiles associated with flooding, transplanting, and growth stages of rice [5,11]. SAR-based approaches are particularly attractive as the SAR signal is not impeded by cloud cover, making them an invaluable resource in most tropical rice-growing areas [12–14]. Moreover, the SAR backscattering coefficient typically exhibits a V-shaped temporal variation, reflecting the sequential changes from bare soil [high] to flooded conditions [low] and to rice canopy [high] [10,15]. Although historical access to SAR datasets was limited due to high cost and long revisit times, the European Space Agency's Sentinel-1 observations allowed a wide range of applications, including rice mapping in many regions around the world [11,16–18].

Previous attempts to map rice-planted areas from remote sensing also include the use of optical observations, although many of these studies rely on high temporal observations provided by sensors like Moderate-Resolution Imaging Spectroradiometer (MODIS) to address the cloud issues, particularly in Monsoon Asia [5,7,19–23]. For example, early work by Xiao et al. (2006) [5] used MODIS observations and a unique combination of Enhanced Vegetation Index (EVI) and Land Surface Water Index (LSWI) time series data to identify rice across South and Southeast Asia. Similarly, Ref. [24] developed a seasonal rice mapping algorithm for Bangladesh using MODIS observations and a decision tree algorithm. More recently, Ref. [23] developed a ~20-year [2000–2022] paddy rice and cropping intensity dataset in Monsoon Asia at 500 m resolution that allowed improved understanding of rice expansion in the region. Despite the success of these studies in locating rice-planted areas, questions remain as to how effective these coarse-resolution datasets are in identifying individual rice fields that dot the landscapes of many rice-growing areas. Rice fields often occur in small patches in fragmented landscapes, particularly in Sri Lanka, rendering coarser-resolution (>100 m) observations insufficient for accurate mapping. While SAR observations have been particularly suitable for mapping rice-planted areas in monsoon regions, limitations in their long-term availability have hindered their usefulness in understanding the changes in rice-planted areas. Moreover, field-scale rice yield estimation, as intended in this paper, is not easily developed from SAR or coarser-resolution data. To address these issues, optical observations provided by MODIS, Landsat, and Sentinel-2 platforms have been proposed [25–30]. For example, Ref. [31] developed a Landsat-based rice mapping index to identify rice-planted areas in Sri Lanka, even though only a portion of the country's rice-planted areas were targeted.

The purpose of this study is to expand on previous remote sensing-based rice mapping efforts to develop a simple yet effective method to derive field-scale rice-planted areas in Sri Lanka using medium resolution (10–30 m) optical remote sensing data for two decades [2000–2022] and limited amounts of training data. More specifically, our rice mapping utilizes a phenology-tuned, rule-based classification algorithm that has been optimized for consistent long-term monitoring, rather than a generic machine learning model. We also developed a field-scale rice yield estimation tool based on a random forest regression algorithm that integrates satellite observations of a chlorophyll index and climatic variables that are deemed to be important for yield assessments and allows us to downscale reported rice production statistics at the regional level down to individual fields.

This study positions itself at the forefront of agricultural monitoring by leveraging the unique capabilities of optical remote sensing to overcome limitations inherent in traditional rice surveys, particularly for long-term time series analysis in cloud-prone, smallholder-dominated regions like Sri Lanka. By providing a consistent, spatially explicit, and historically rich dataset, our approach offers a powerful tool for developing highly targeted interventions, assessing long-term food security trends, and supporting climate-resilient agricultural policies, with direct applicability to similar challenging agricultural contexts in Southeast Asia and beyond.

Our study also provides an opportunity to examine the effects of Sri Lanka's 2021 ban of synthetic fertilizers on crop yields. More specifically, observed inter-annual variability driven by climate and other factors, including those from a fertilizer ban, is captured by the remote sensing data. For example, our findings suggest that rice yields were reduced by more than 30 percent in the year the ban was introduced. The work presented here provides the foundational dataset for studies that could employ statistical methods (e.g., difference-in-differences and regression models with control variables) to investigate the impact of specific interventions or confounding factors on rice yield but in a more spatially explicit manner, which would not have been possible to examine with aggregate government statistics.

2. Materials and Methods

2.1. Description of the Study Area

Sri Lanka is a tropical island nation located in the Indian Ocean, off the southern tip of the Indian subcontinent. It has a land area of about 66,000 km^2 that ranges from sea level to over 2500 m above sea level in central parts of the country. As a tropical country with a significant population dependent on rice production and a favorable climate for multiple crops per year, Sri Lanka produces at least two rice crops per year, spread over 950,000 hectares of land (Figure 1). The first crop is produced during the Maha (major) season from September to March. The second crop is produced during the Yala (minor) season from March to August [4]. Rice is grown across all climatic zones of the country, but the low-lying plains in the east, northwest, and south dominate cultivation within government-supported projects of various sizes and production volumes. Sri Lanka has large areas of irrigated cultivation, many of which have had a well-distributed cascade irrigation network since colonial times. These irrigation schemes provide water for rice cultivation in both seasons, especially in the dry zones (depending on seasonal water availability).

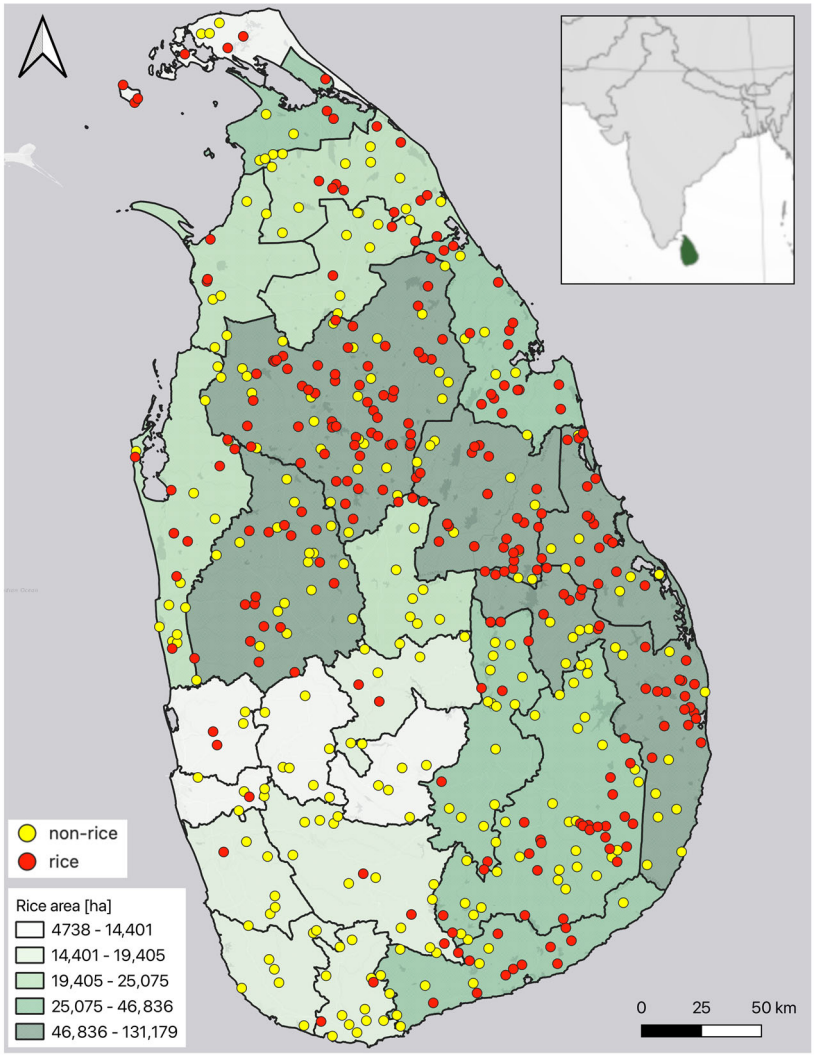


Figure 1. Map of Sri Lanka showing district boundaries along with rice-planted areas in 2023. Also shown are the locations of stratified random samples drawn for rice (red) and the non-rice (yellow) categories used in accuracy assessment for 2021 for the Maha season. The rice-planted areas are from Sri Lanka's Department of Census and Statistics [4]. The inset map showing the location of Sri Lanka in Southeast Asia was acquired from Wikipedia under the CC BY-SA 3.0 permission [https://commons.wikimedia.org/w/index.php?curid=5570766, URL accessed on 8 August 2025].

2.2. Data Acquisition

In this study, we used two primary types of data: (1) remotely sensed observations in the optical domain from Landsat and Sentinel-2 and (2) paddy rice statistics available from the government of Sri Lanka [4]. The selection of Landsat and Sentinel-2 satellite data for rice area and yield mapping in Sri Lanka is fundamentally justified by their complementary strengths, which collectively address the unique challenges of agricultural monitoring in cloud-prone, smallholder-dominated regions. In particular, Landsat imagery, with its unparalleled archive extending back to the 1970s, provides a crucial historical perspective. Its consistent 30 m spatial resolution across multiple spectral bands is well-suited for capturing the spatial patterns of rice fields, even those belonging to smallholder farmers, which are often too fragmented to be accurately resolved by coarser-resolution sensors like MODIS. Similarly, Sentinel-2, a more recent mission, significantly augments Landsat's capabilities. Its higher spatial resolution (10–20 m for key bands) provides even finer detail, crucial for distinguishing small, irregularly shaped rice paddies and capturing within-field

variability. More importantly, Sentinel-2's high temporal resolution, with a revisit time of approximately 5 days (and even more frequent with multiple satellites and overlapping orbits), dramatically increases the probability of acquiring cloud-free observations during critical rice phenological stages.

Remotely sensed observations were accessed and processed within Google Earth Engine (GEE), a large-scale geospatial platform [32]. All available Landsat Thematic Mapper (TM), Enhanced Thematic Mapper Plus (ETM+), and Operational Land Imager (OLI) observations from Collection 2 processing Level 2 between 2000 and 2023 with less than 50% cloud cover across Sri Lanka were requested from the GEE archive in the form of surface reflectance, surface temperature, and quality assurance (QA) products. It should be noted that the 50% threshold was selected as a pragmatic balance to maximize the availability of usable, cloud-free pixels for time series analysis to ensure sufficient data points for accurate phenological curve reconstruction. All images were cloud-screened using the cloud and cloud shadow masks separately that were available in the QA bands [33]. Landsat observations were supplemented with Sentinel-2 surface reflectance observations starting in 2019 because high-quality and atmospherically corrected observations are only available starting in that year. Sentinel-2 surface reflectance observations were cloud-screened using the Cloud Score+ QA processor outputs that were available for each scene [34]. Sentinel-2 observations were matched to Landsat in terms of spatial resolution and placed in the time series stack. No band-pass match or other corrections were performed other than a spatial resolution match (to Landsat). Overall, the number of cloud-free images ranged from a few to over 200, depending on the year and sensor (Figure 2).

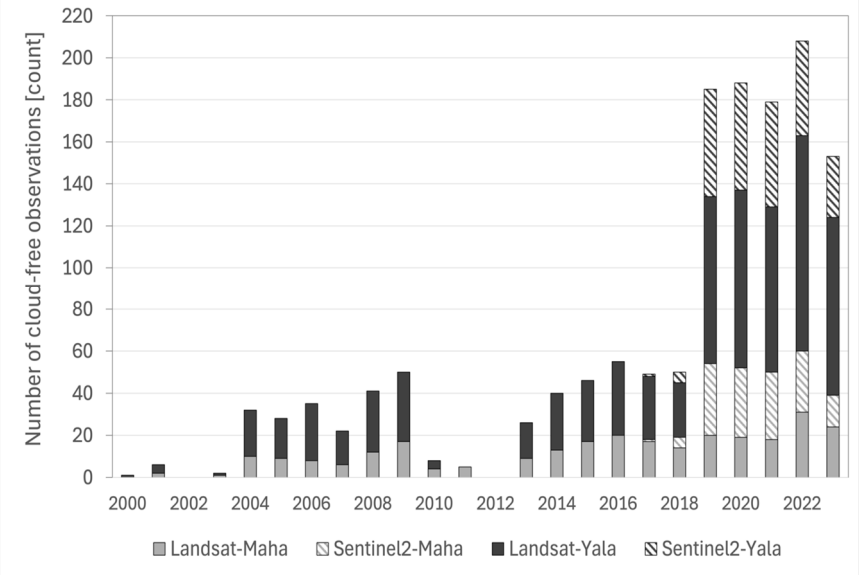


Figure 2. Number of cloud-free observations across Sri Lanka by year and by sensor. A minimum of 20% percent cloud cover filter is applied.

The paddy statistics data was provided by the Sri Lankan government [4]. These datasets were in the form of district-level paddy rice area and yield by season (both Maha and Yala), as were the original crop cutting studies that were used to develop the rice statistics in the country. These datasets were used to evaluate the quality of the rice area maps and to build the rice yield model described in the sections below. It should be noted that, while the crop cutting experiments are conducted in locations selected at random and stratified by production intensity [4], we did not have access to location information for site-specific validation, so they were only used to construct yield histograms, which are then compared to those derived from remote sensing.

In addition to satellite observations, we also relied on several ancillary datasets to isolate the rice crop signal. We use a land cover mask derived from WorldCover [35] and an urban area mask derived from the Global Human Settlement [36] product (Table 1). These datasets were useful in removing occasional false positives, defined as detecting fields as rice although no rice is present.

Table 1. Variables used in this study.

Variable	Source	Spatial Resolution
NDVI	Landsat/Sentinel-2	10–30 m
LSWI	Landsat/Sentinel-2	10–30 m
WorldCover	Sentinel	10 m
Global Human Settlement	Sentinel	10 m
GCI	Landsat/Sentinel-2	10–30 m
Seasonal rain [mm]	ERA5	10 km
Rainy days	ERA5	10 km
Vapor Pressure Deficit (VPD)	ERA5	10 km
Days without rain	ERA5	10 km
Days w/temp > 30 °C	ERA5	10 km
Elevation	SRTM	30 m
District boundary	GADM	30 m [rasterized]

2.3. Paddy Rice Area Mapping Method

Remote sensing, either alone or in combination with ground-based statistical surveys, has been used to identify and map paddy rice areas for over three decades [4,6,9,15,20,31,37–39]. The advantages of a remote sensing-based solution include (a) an objective assessment; (b) a synoptic view covering a large area; (c) an archival nature of data, allowing changes to be monitored; and (d) the advantage of established methods that are robust and tested for identifying croplands and their irrigation status [40].

The presence of rice-growing areas is determined by a combination of factors, including climate, water availability, topographic position, farmer decision-making, and technical expertise. To this end, the methodological approach followed in this study is centered around monitoring rice-related vegetation and water management activity from space and distinguishing this activity from all other land use types using both spectral and temporal information contained in the satellite signal (Figure 3).

To identify and map rice-planted areas in Sri Lanka, we use time series observations from Landsat and Sentinel-2 sensors that allow us to monitor fields as small as 0.25 hectares. The approach described here produces crop type and yield estimates at the pixel level, and it would be possible in principle to aggregate these pixels to produce field-level estimates. However, we did not have access to a field boundary dataset that would allow this aggregation. Recent work involving field boundary detection in satellite imagery across large areas [41] would greatly help in this regard.

The approach taken here is based on maximizing the rice-related information present in spectral bands or indices derived from remotely sensed observations and the ability to relate this information to complex forms of rice occurrence across the country. Two such indices are related to *greenness* and *wetness*. A schematic of how these indices can be used to identify paddy rice in Sri Lanka is shown in Figure 4, in which a group of paddy rice fields located to the east of Hettipola exhibits a distinct vegetation and water temporal cycle that is characteristic of rice growth.

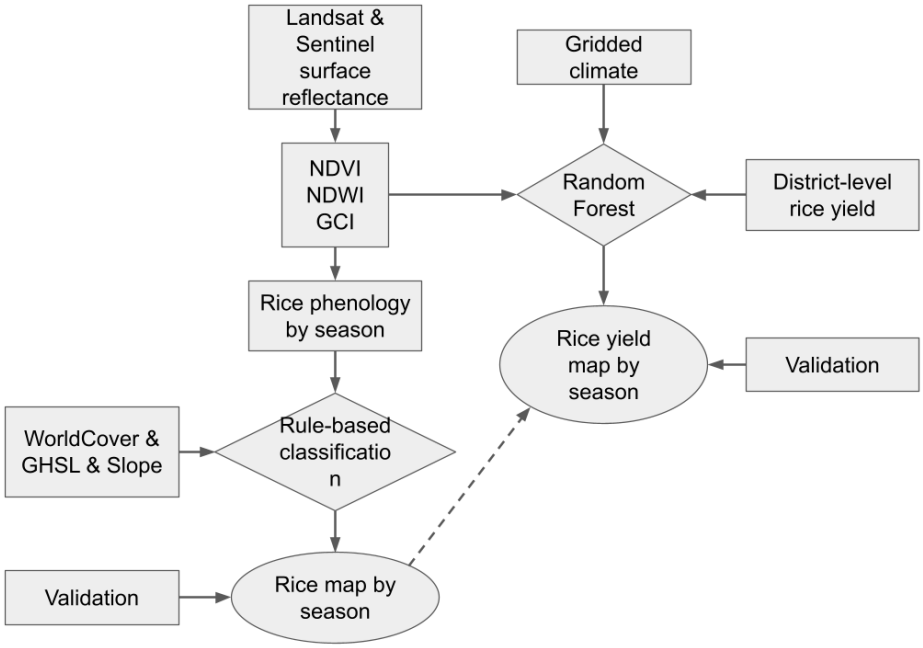


Figure 3. Methodological flowchart summarizing our approach taken in this study to map rice-planted areas and their yields. Refer to Table 1 for details on the datasets used.

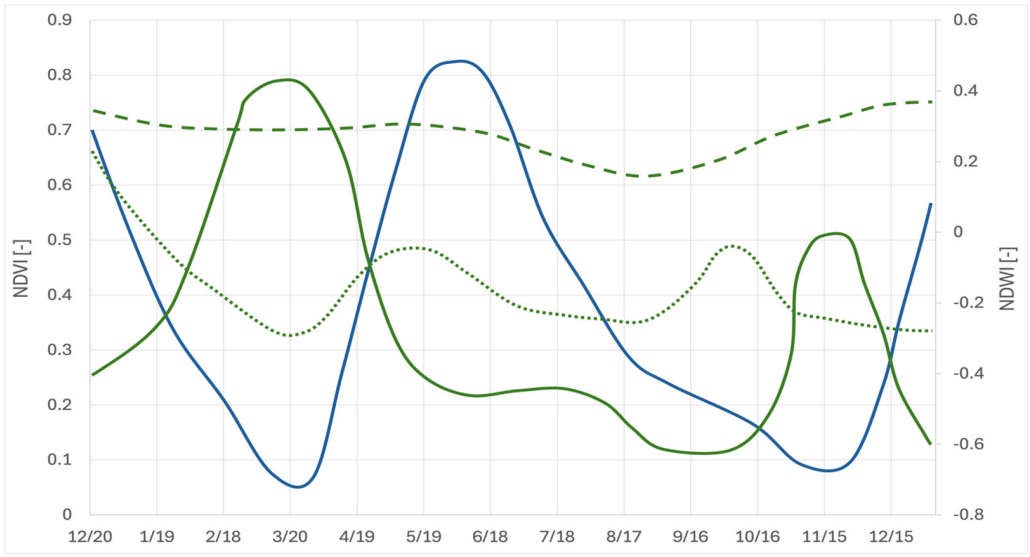


Figure 4. Idealized (smoothed) temporal growth trajectory for three land cover types captured from remotely sensed observations in the form of wetness (blue) [right vertical axis] and greenness (green) [left vertical axis] around the town of Hettipola in Sri Lanka. The solid line is for paddy rice, the dashed line is for tropical forest, and the dotted line is for grassland. Of particular importance is the contrast between the wet (transplanted) and green periods of rice that can be captured and used to map rice presence/absence in the country.

More specifically, when rice is transplanted into paddy areas, fields show a strong water signal evidenced by a large wetness index (solid blue line in Figure 4). As the rice plant grows, the water signal gradually gives way to the vegetation signal associated with emergence and growth, as indicated by the solid green line in Figure 4. When contrasted with other land cover types, namely forests and grasslands, this signature is unique to paddy rice and can be used to construct a rice index. The main message from Figure 4 is that paddy rice has distinct vegetative and wetness signatures associated with rice cultivation

practices, and these signatures can be automatically captured and used to identify and map rice areas regardless of their size.

To employ the proposed approach, we used spectral indices that have been shown to be sensitive to rice detection in previous studies [5,42]. More specifically, we used the Normalized Difference Vegetation Index (NDVI) and the Normalized Difference Water Index (NDWI), defined as follows:

$$NDVI = \frac{\rho_{nir} - \rho_{red}}{\rho_{nir} + \rho_{red}} \quad (1)$$

$$NDWI = \frac{\rho_{green} - \rho_{nir}}{\rho_{green} + \rho_{nir}} \quad (2)$$

where ρ_{band} refers to surface reflectance in the respective green, red, and nir (near-infrared) spectral bands. One of the challenges with using these indices to identify paddy rice fields in Sri Lanka was using observations from Landsat/Sentinel-2 with relatively low temporal frequency and irregular availability. We addressed this by extracting the magnitude and timing of maximum vegetation and wetness signals from the time series observations and encoding them into phenologically derived rice index expressions (Equations (3)–(5)). For example, in the Maha season, several rice indices were identified as follows:

$$\boldsymbol{R_1}[\boldsymbol{rice}] = \mathbf{2} - (\boldsymbol{NDVI_{95}}[\boldsymbol{Maha}]) + (-\mathbf{1} \times \boldsymbol{NDWI_{95}}[\boldsymbol{Maha}]) \quad (3)$$

$$\boldsymbol{R_2}[\boldsymbol{rice}] = (\boldsymbol{NDWI_{95}}[\boldsymbol{Maha}] + \mathbf{1}) - \boldsymbol{NDVI_5}[\boldsymbol{Maha}] \quad (4)$$

$$\boldsymbol{R_3}[\boldsymbol{rice}] = \boldsymbol{NDVI_{95}}[\boldsymbol{Maha}] - \boldsymbol{NDVI_5}[\boldsymbol{Maha}] \quad (5)$$

where R_x[rice] refers to the rice index, and $NDXX_{95}$ and $NDXX_5$ refer to 95th and 5th percentiles of the spectral indices within the season, respectively. The purpose of R_1 is to enhance the vegetation signal at peak rice growth period when NDVI is large and contrast it with the NDWI signal at max at transplanting (Figure 4). The utility of R_2 is to enhance the water signal associated with transplanting early in the season (Figure 4). The purpose of R_3 is to ensure that the signal is associated with vegetation activity typical of crops with lower (planting/transplanting), followed by higher (peak growth), and followed by lower (senescence) vegetation index values. We then add all three indices together and apply a rule-based classification algorithm that utilizes an automated threshold selection process to separate rice from non-rice crops and other land cover categories. More specifically, we use a threshold developed from the histogram of summed rice index values as illustrated in Figure 5, which shows the stacked histograms of three land cover types, including rice, non-rice croplands, and other land cover types, including forests and grasslands. There is a clear distinction between the concentration of the proposed index values for rice (far right) that could be separated using a threshold value of about 1.6 (Figure 5). We use the histogram minima approach, which refers to the valleys or local minima between peaks in a histogram and allows us to separate data points into distinct groups or classes, and including rice as the minimum point between the peaks is the most logical choice for a threshold. It should be noted that there is some overlap between rice and non-rice crops and, to a smaller degree, with other land cover categories, and therefore, the use of a single index may not always be able to capture all rice-planted fields. However, accuracy assessment (described in detail below) suggests that these errors are at acceptable levels (less than 10 percent).

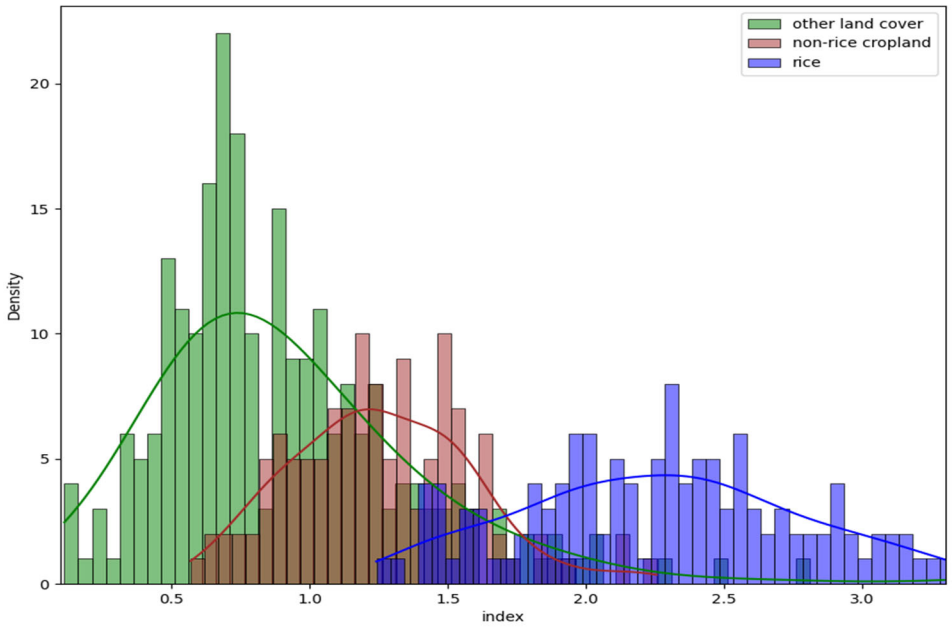


Figure 5. Stacked histogram distribution of summed rice index values [$R_1 + R_2 + R_3$] for three land cover categories, including rice (blue bars), non-rice croplands including maize and other crops (brown bars), and other land cover categories such as forest and grasslands (green bars) in the Maha 2020 season. Also plotted are the estimated kernel densities derived from underlying data as solid lines. The histograms were developed from roughly 60 samples placed in each group.

The summed R indices show strong spatial patterns of rice presence, from which binary rice/non-rice pixel labels are drawn. Several examples of the summed R index from various parts of Sri Lanka under different climatic and growing conditions are provided in Figure 6. It should be noted that the proposed method relies on the presence of high-quality forest and water (river/lake/reservoir/ocean) masks, derived from remotely sensed observations as listed in Table 1.

To produce pixel-level binary rice maps, we applied a dynamic thresholding approach applied to histogram minima separately for the Maha and Yala seasons for each year between 2000 and 2022 for all of Sri Lanka.

2.4. Paddy Yield Mapping

To develop paddy rice yield maps, we relied on the relationship between district-level rice yields provided by the government [4] and the remote sensing-based green chlorophyll index (GCI) put forth by [43] for pixels identified as rice in the previous section. The GCI is calculated as follows:

$$GCI = \frac{\rho_{nir}}{\rho_{green}} - 1 \quad (6)$$

where ρ_{band} is defined earlier. The GCI has been shown to be highly correlated with crop yields in many environments [44,45]. Using this approach, the paddy rice yield mapping problem is reduced to a downscaling problem in which a regression model trained with the seasonal (both Maha and Yala) maximum value of GCI, defined as the 95th percentile or GCI_{95}, and reported yield data at the district level is applied to make field-scale paddy rice yield predictions. The GCI_{95} (or the maximum value of GCI during the growing season) is typically reached during the panicle initiation stage. We identified this critical stage for each rice season based on satellite observations indicating the period of maximum biomass accumulation. This selection is crucial as GCI at this stage is highly correlated with biomass and, consequently, final yield. It should be noted that the use of GCI_{95} has the additional benefit of accounting for phenological variations across different sub-regions within our

study area. More specifically, our method incorporates a dynamic approach to identify the peak GCI for each individual pixel or sub-region based on its specific time series profile. This ensures that the GCI value used for yield estimation is consistently captured at the most relevant physiological stage for each specific rice-growing parcel.

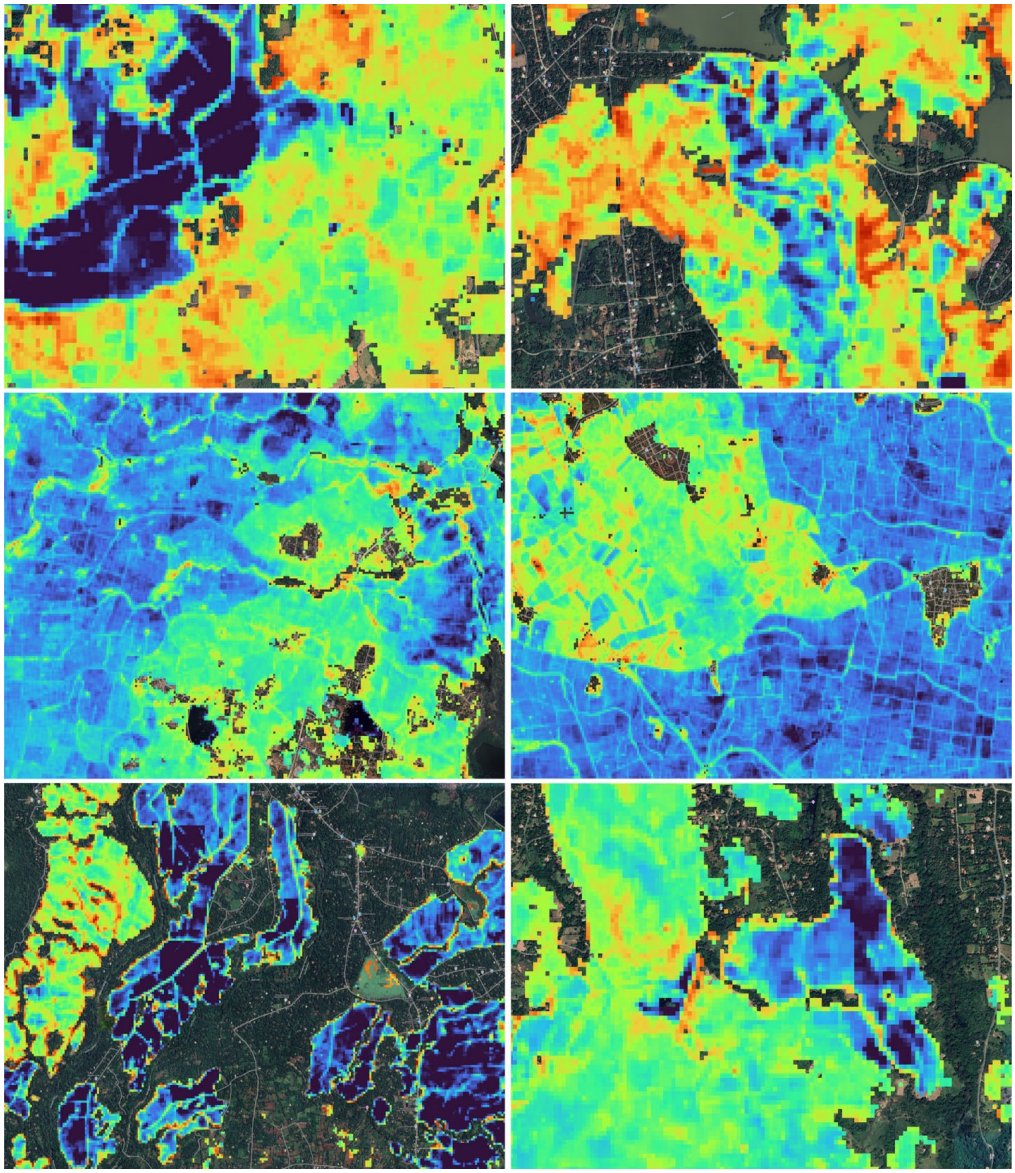


Figure 6. Spatial patterns of the summed rice index from various parts of Sri Lanka. In general, hues of dark blue indicate a high rice index associated with rice presence, while colors of green, yellow, and red indicate the presence of non-rice crops and other land cover types. Attention is drawn to examples with a large contrast of rice fields (dark blue hues) juxtaposed against non-rice areas (green to yellow to red hues).

In addition to the GCI_{95} variable that has been shown to be strongly related to crop yields, variables describing the environmental conditions in which rice is cultivated further improve remote sensing-based yield estimation efforts [44,45]. To this end, we included additional environmental variables listed in Table 1 and opted for a machine learning algorithm-based regression model. More specifically, we used a random forest regression algorithm to produce pixel (field)-level yields. The environmental variables, particularly those representing climate conditions, were selected based on previous work on developing high-resolution yield datasets in Asia and elsewhere [25,44] and using expert judgment,

balancing our knowledge on rice agronomy against data availability for climate variables in existing gridded sources. It should be noted that we did not include variables related to management practices such as fertilizer applications or irrigation applications, or soil type, as these variables were not available at the granular scale needed. As such, remotely sensed observations along with environmental conditions were sufficient to capture field-scale yield variations that were of interest in this study [25], and management practices were not needed to derive good yield estimates. All data were resampled to a 30 m spatial resolution and prepared as a multi-layer (band) stack to be used in random forest regression estimates.

Random forest regression is a supervised learning algorithm that uses an ensemble learning method for regression in machine learning [46]. Our implementation of the algorithm includes a Bagging (or Bootstrap Aggregations) approach that allows the trees in random forests to run in parallel, meaning there is no interaction between these trees while building the trees. The only hyperparameter setting was for the number of trees, which was set at 150 based on our prior work [44].

To account for the district effect, we also applied one-hot encoding to districts to improve prediction accuracy. With one-hot encoding, each district's code is converted into a new categorical column and assigned a binary value of 1 or 0. The one-hot-encoding approach allowed us to remove the ordinality issue between the districts since most machine learning algorithms treat the order of numbers as an attribute of significance.

It should be noted that of the eight predictor variables used in yield modeling with a random forest regression, only the GCI and elevation variables provide the spatial resolution needed to produce field-scale quantities. While there may be some concern about the effects of the coarse-resolution nature of the climatic variables, the results do not support this concern, and field-level granularity is retained in the final yield maps as expected.

The application of the random forest model to district-level data suggests high prediction accuracy, based on 10-fold cross-validation in which roughly 25 percent of the training data was held for testing purposes (Figure 7). It should be noted that we did not have access to field-scale yield data for validation purposes and used the test hold-out approach to examine how well the district-level yield quantities are reproduced by the random forest regression model. On the other hand, we had access to data from national crop cutting experiments without individual plot locations. These data were used to assess the statistical distribution of yield estimates as described below.

The random forest model was applied to all pixels judged to be cultivated with rice to produce nationwide yield data at the pixel (field) level for each year since the year 2000, separately for the Maha and Yala seasons.

2.5. Accuracy Assessment

The accuracy assessment (or validation) of the seasonal rice maps derived from satellite data was conducted to quantify how reliable each map is and how the errors are distributed by location and category. To accomplish this accuracy assessment part, established protocols [47] were used. First, a stratified random sample in which samples are stratified based on class type (rice and non-rice) was drawn from the final maps and evaluated against their true label. For each sample, the true category was identified using all available information. In most cases, we relied on high-resolution images available through Google Earth, where images were used as a substitute for ground reference, although these images may be snapshots of rice fields and irrigation status taken at different times. Nevertheless, the high-resolution datasets used in accuracy assessment had the advantage of being comprehensive, providing information on much larger areas, and, therefore, being representative of the entire country. When available, we also used Planet high-resolution data made available

through Norway's International Climate and Forests Initiative (NICFI) program [48]. The geographic distribution of these samples in the Maha seasons for a single year is provided in Figure 1. In each season (Maha and Yala) and year, the size of the reference sample varied but was typically between 200 and 250 individual points. When labeling the test sites as their true category, the decisions were often based on shapes, topographic position, water presence, proximity to a water source, and the size of individual paddy fields. It should be noted that due to the remote collection nature of the test data, it is possible that some cases were incorrectly labeled. At this time, the error rate in the test data is unknown but is assumed to be small (less than 5% would be a good guess), and therefore, our validation approach does not show this to be a major concern.

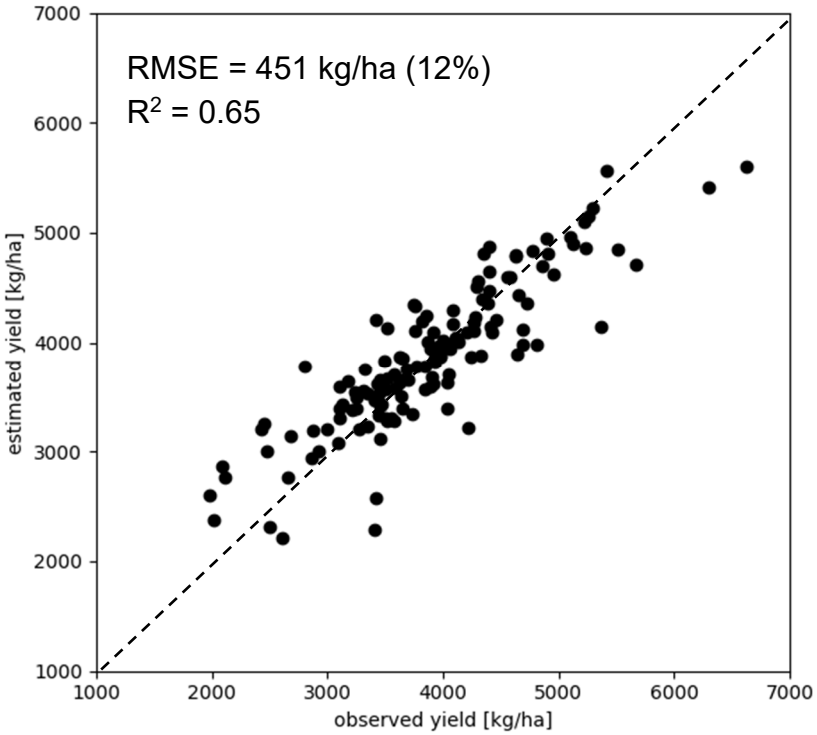


Figure 7. District-level validation results of the random forest regression model trained with remotely sensed observation and environmental variables, with district yields from [4]. The validation is based on roughly 25 percent of the training data held back for testing purposes in a cross-validation. Each data point represents a district in the test set and represents the mean of 10-fold cross-validation.

The overall and marginal class-specific accuracies (or errors) were computed with the help of a confusion matrix—a standard tool for assessing the accuracy of land cover and land use classifications. In a standard confusion or error matrix, the rows correspond to true labels (the test set) while the columns correspond to labels predicted in the final rice maps. The diagonal elements in the matrix represent the number of correctly classified samples of each class, i.e., the number of ground reference samples of a certain category that obtained the same class label in the analysis. In contrast, the off-diagonal elements represent misclassified samples or the analysis errors, i.e., the number of reference samples that ended up in another category during analysis. Such errors are also known as errors of omission or exclusion and as errors of commission or inclusion. It should be noted that the sample-based estimates of map accuracy are statistical estimates—information about the total population (i.e., all pixels in the map) is inferred from a sub-population (i.e., the sampled pixels). The stratified random sampling approach allowed us to overcome this issue by precisely allocating the number of desired samples within each class (or region) of interest.

Finally, we adjusted the paddy rice areas derived from remotely sensed observations using the confusion matrix following [49]. The confusion matrix provided the basis for these adjustments as it allows us to quantify omission (e.g., missing rice pixels) and commission (e.g., extra rice pixels) errors and therefore correct rice area estimates upward or downward, respectively. It should be noted that because of the roughly equal amounts of omission and commission errors present in the seasonal rice maps (see Section 3.1 below), the area adjustment did not result in major changes in original areas derived directly from the maps.

The second form of validation was conducted by comparing paddy areas derived from remote sensing to those produced by the Sri Lankan government [4]. To achieve this, we first multiplied each pixel identified as rice with its respective area (900 square meters for 30 m pixels) using an equal area projection map of rice and then summed all the pixels within each district and converted to hectares. These comparisons were made for each season at the district level, which is the only subnational reporting unit for paddy statistics [4].

3. Results

3.1. Paddy Rice Map Accuracy

The overall accuracy of the rice map for the 2020/2021 Maha season is about 91% and for the 2021 Yala season, it is about 93% (Tables 2 and 3). Class-specific omission and commission errors for the year 2021 paddy rice map for Sri Lanka are provided in Table 4.

Table 2. Confusion matrix associated with the national Maha paddy rice map of 2021 derived from remotely sensed observations. The samples were derived from a stratified sample of roughly 500 points, stratified on the final paddy rice map. The geographic distribution of samples for the Maha season is given in Figure 1.

		Reference		
		non-rice	rice	total
Map	non-rice	222	27	249
	rice	19	232	251
	total	241	259	500

Table 3. Confusion matrix associated with the national Yala paddy rice map of 2021 derived from remotely sensed observations. The samples were derived from a stratified sample of roughly 500 points, stratified on the final paddy rice map.

		Reference		
		non-rice	rice	total
Map	non-rice	235	17	252
	rice	20	228	248
	total	255	245	500

It should be noted that while rice maps were produced for the 23-year period (2000–2022), accuracy assessments have only been carried out for the last seven years (2016–2023), with the assumption that years prior to 2016 have similar error characteristics. However, given that the Landsat archive contains a lesser number of images prior to 2013, when Landsat 8 was launched, it was possible that the resulting rice maps prior to 2013 were not as reliable. Nevertheless, the accuracy assessment reveals a pattern of roughly equal amounts of omission and commission errors across the two categories. The Yala 2021 paddy rice map is slightly better in terms of overall as well as omission and commission

errors, potentially because of lower cloud cover that affects all optical remote sensing mapping efforts in the tropics.

Table 4. Class-specific errors of omission and commission in both seasons for the 2020/2021 growing season.

Year/Season	Category	Error [%]
Maha 2021	Non-rice omission	7.9
Maha 2021	Rice omission	10.4
Maha 2021	Non-rice commission	10.8
Maha 2021	Rice commission	7.6
Yala 2021	Non-rice omission	7.8
Yala 2021	Rice omission	6.9
Yala 2021	Non-rice commission	6.7
Yala 2021	Rice commission	8.1

Understanding where errors occur geographically is crucial for improving rice maps' utility, as errors may not be uniformly distributed across Sri Lanka, and certain regions or districts might exhibit higher or lower error rates. One way to quantify the geographic distribution of map errors is to analyze the confusion matrix at the district level (Figure 8). Ideally, such an effort would be focused on producing separate stratified random samples in each district. However, given the tremendous effort to produce true labels manually, we used the national sampling framework described earlier and simply allocated the samples across districts based on their rice-cultivated area. As a result, there is the possibility that some districts, especially those with smaller areas of rice cultivation, may not have received enough samples in each district to draw meaningful conclusions. With that note, this analysis reveals that, in general, those districts with larger areas of paddy rice, especially those along the eastern and southern coasts of Sri Lanka, have higher map accuracies when compared to some of the higher elevation districts in Sri Lanka's hinterlands with lower occurrences of paddy rice fields. This is true for both the Maha and Yala seasons. It should be noted that while western and central districts with more complex topography and landscape elements are expected to have lower accuracies, partly because of the complex arrangement of paddy rice areas in these locations, they also have a smaller sample of validation points due to the sampling strategy that may confound the accuracy metrics.

3.2. Paddy Rice Areas

The remote sensing-based paddy rice identification methodology developed here produced seasonal (Maha and Yala) rice maps for each year between 2000 and 2022 in Sri Lanka. These maps show expected spatial and temporal patterns of rice cultivation across the country, reflecting the complex interplay of environmental, socio-economic, and technological factors (Figure 9). The utilization of remote sensing data and classification algorithms facilitates a comprehensive understanding of these patterns with unprecedented granularity (Figure 10). Geographically, Sri Lanka's rice cultivation is predominantly concentrated in the lowland areas, particularly in the northern, eastern, and north-central districts, where the monsoon rainfall pattern facilitates wet rice farming. These districts have historically been the rice bowl of the country. Remote sensing data allows for precise mapping of these areas while tracking changes over time. Notably, in recent years, there has been an expansion of rice cultivation into previously non-traditional areas, possibly due to improved irrigation infrastructure, promoting the spread of rice fields to the central as well as more southern Badulla and Monaragala districts.

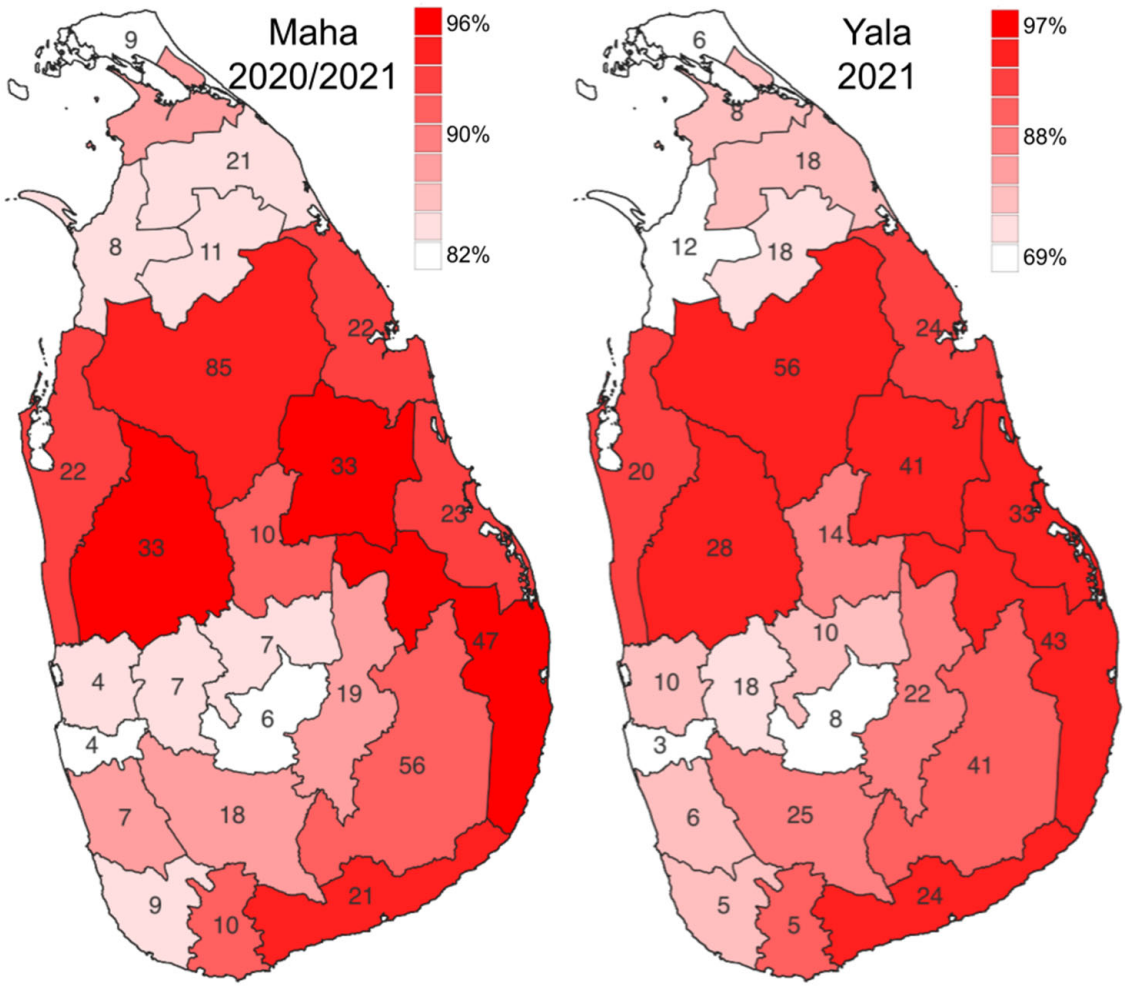


Figure 8. District-level paddy rice map accuracies for the Maha (**left**) and Yala (**right**) seasons in 2021. Colors in the map represent the overall map accuracy in each district, while the numbers in districts represent sample size per district, allocated from the national sampling strategy described in the text and displayed in Figure 8.

At the national level, the remote sensing-based area estimates closely track the areas estimated from survey-based crop cutting experiments [4] for both seasons (Figure 11a,b). The Yala season rice area estimates from remote sensing appear to be better correlated with the statistical estimates from the government, in line with the accuracy estimates described earlier.

District-level comparisons (Figure 12) follow national estimates with one difference. There appears to be higher divergence in rice area estimates in districts with smaller paddy areas (typically less than 20,000 hectares) than those with larger paddy areas. This finding is also in line with district-level accuracy estimates described in Section 3.1.

3.3. Paddy Rice Yields

The application of the random forest machine learning model reveals important patterns of rice yield across Sri Lanka. First, rice grain yield estimates range from 1700 to 7300 kg/ha, which are in line with reported figures for rice production in the country [4]. The least and most productive regions, mostly concentrated in the western and southern parts of the country, follow expected patterns that are controlled by topography, climatic conditions, and management practices. Second, the regression model produces yield maps at the full spatial resolution of Sentinel-2 and Landsat data (collectively at 30 m) for the

whole country. One example of these maps for the Hambantota district, where yield estimates range from 2000 kg/ha (red colors) to a maximum of 7500 kg/ha (green colors), is provided in Figure 13. The more productive areas are located in the central portions of the district, whose morphological characteristics (flat and expansive areas of paddy rice farming) determine a greater amount of yield. Of note is the almost uniform distribution of yields in 2021 but significant declines in 2022, possibly due to the fertilizer import ban.

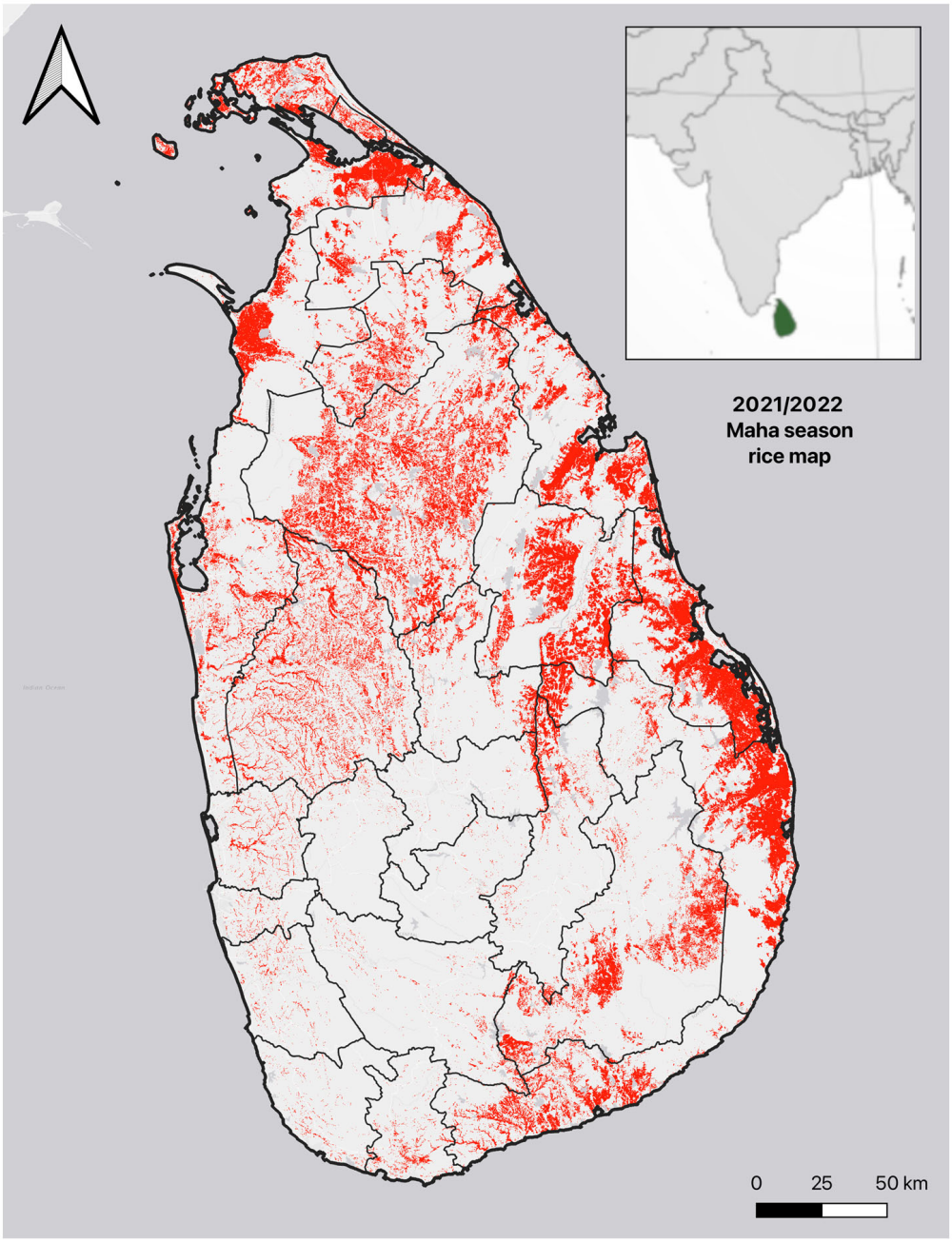


Figure 9. Maha 2021/2022 rice map of Sri Lanka derived from remote sensing. The red-colored locations are individual image pixels (fields) identified as rice following the methodology described in the text. Also shown are the district boundaries drawn in lighter black lines. The inset map showing the location of Sri Lanka in Southeast Asia is acquired from Wikipedia under the CC BY-SA 3.0 permission [https://commons.wikimedia.org/w/index.php?curid=5570766, URL accessed on 8 August 2025].

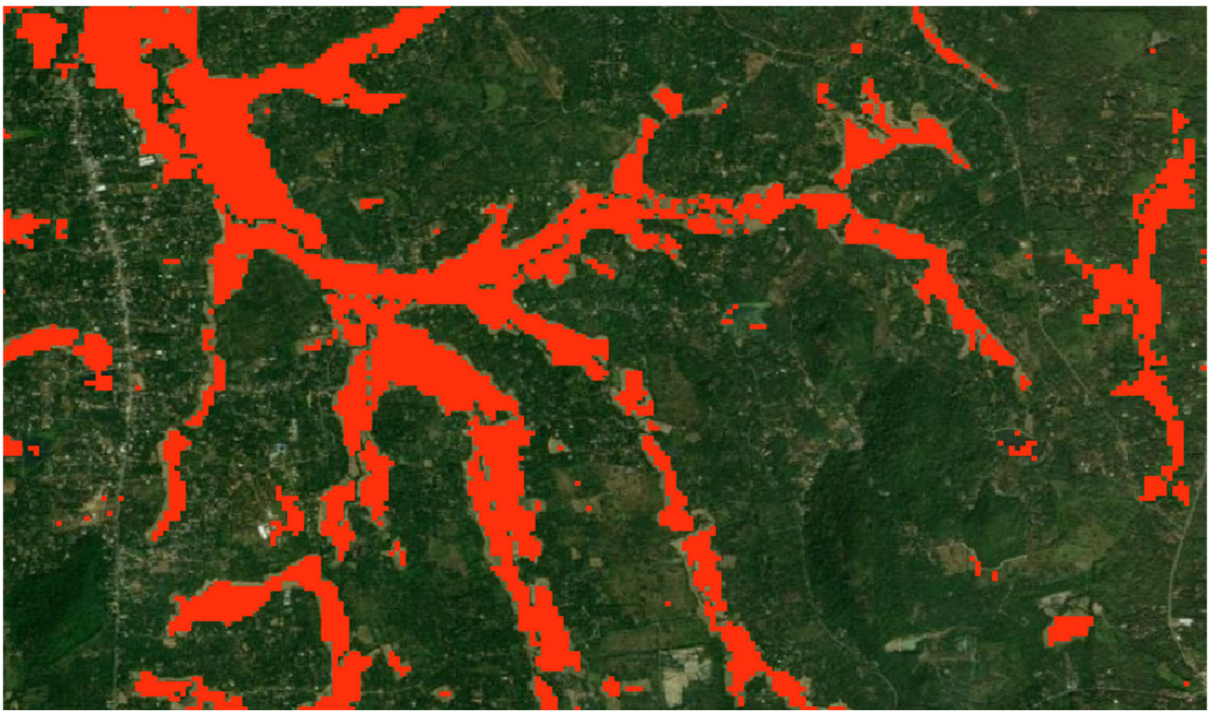
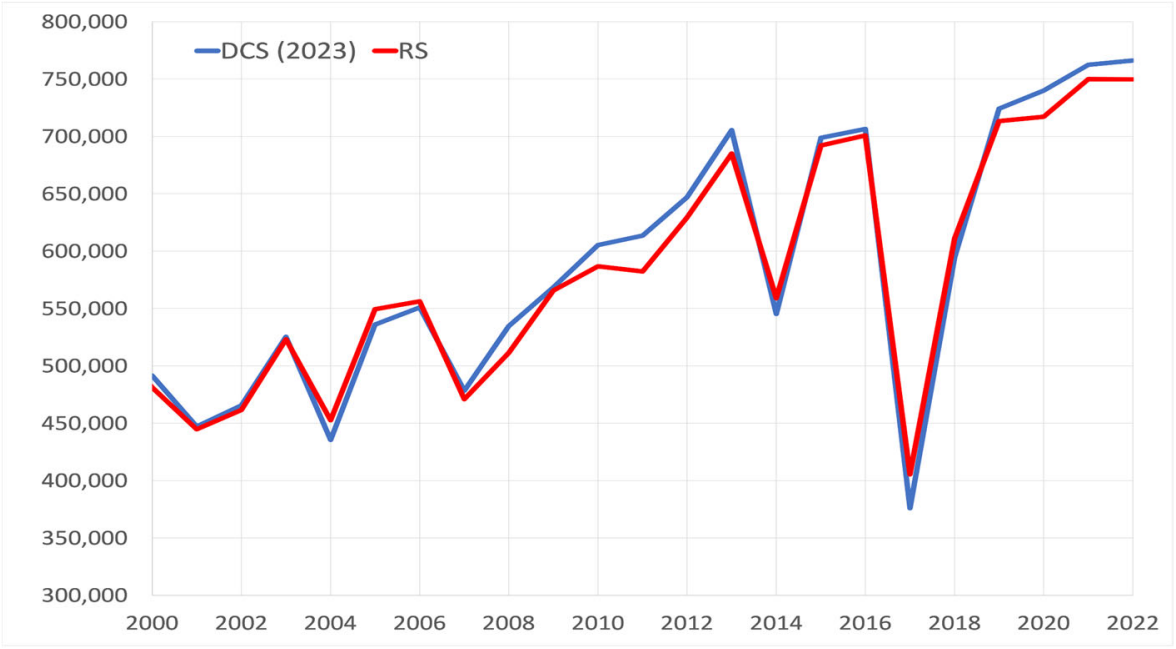
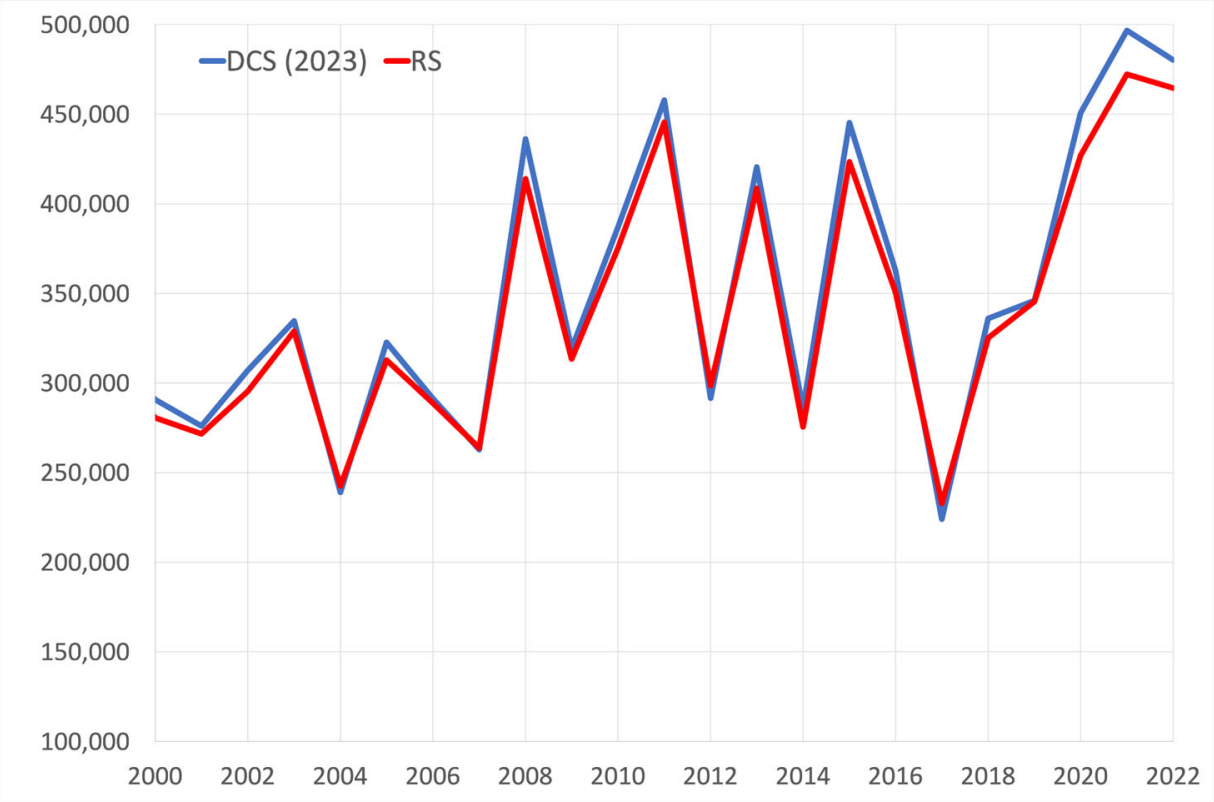


Figure 10. Detailed rice map in western Sri Lanka, around the town of Kurunegala. In this landscape, rice is cultivated in narrow river valleys, following water availability. Even in this complicated landscape, the mapping algorithm works well, capturing rice-cultivated areas.



(**a**)

Figure 11. *Cont.*



(**b**)

Figure 11. (**a**) National rice area statistics derived from remote sensing (RS or red) compared to national crop cutting experiments (DCS or blue) in the Maha season for the last 23 years [2020–2022]. Of note are the years 2021 and 2022 (the last two data points in the chart), when the fertilizer ban took effect, which do not show any decreases in cultivated area. (**b**) National rice area statistics derived from remote sensing (RS or red) compared to national crop cutting experiments (DCS or blue) across a 23-year [2000–2022] window for the Yala season. Of note are the years 2021 and 2022 (the last two data points in the chart), when the fertilizer ban took effect, which show some minor decreases in cultivated area.

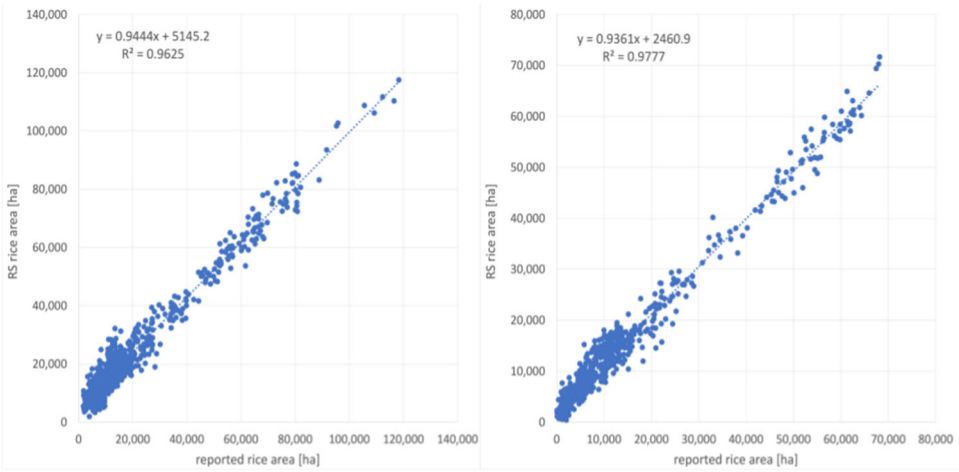


Figure 12. District-level paddy rice area comparisons between reported (DCS, 2023) [4] on the X-axis and remote sensing-based estimates on the Y-axis for the Maha (left panel) and the Yala (right panel) seasons across the 23-year [2000–2022] window. Each point in the plots represents a single district–year combination.

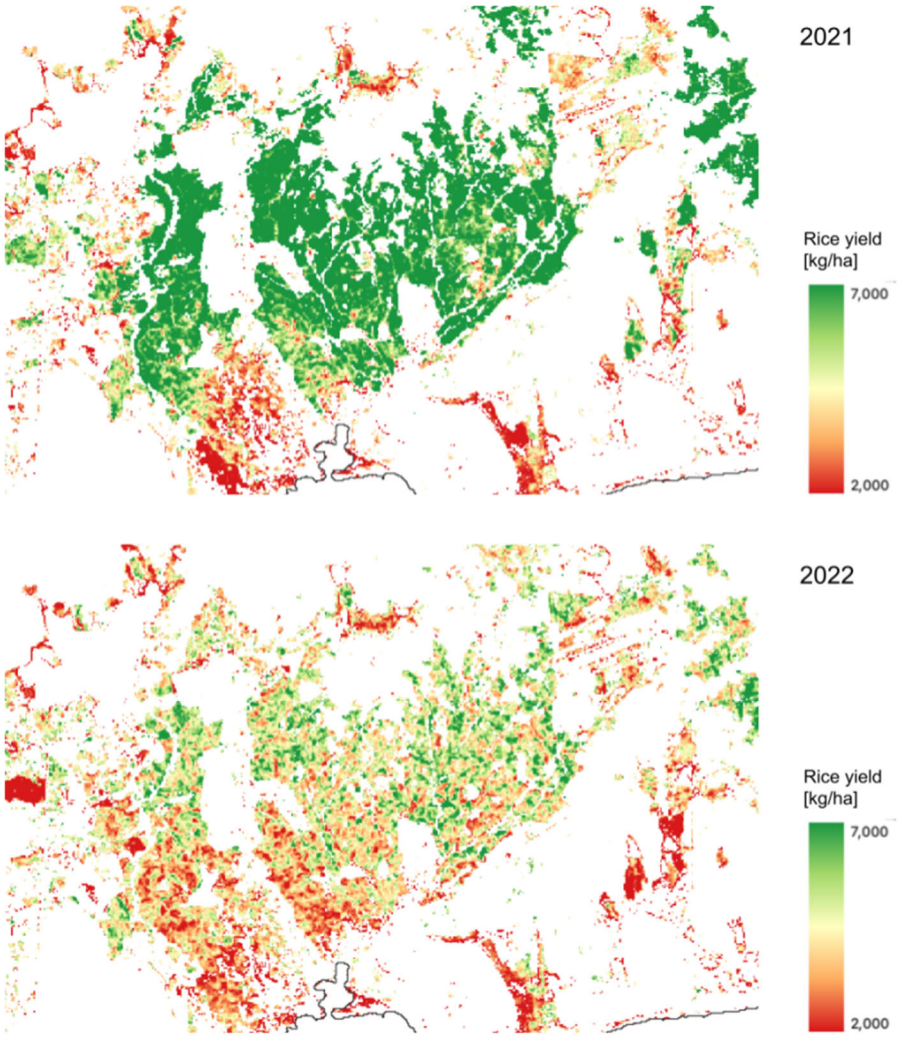


Figure 13. Detailed view of field-level Maha paddy rice yield map for 2021 (top) and 2022 (bottom) in the central part of the Hambantota district (southern Sri Lanka) produced by remote sensing, showing reduced yields. The approximate scale of the map is 1:25,000.

We also compared pixel-level rice yield estimates from the regression model to yield data acquired from crop cutting experiments in eight districts in histogram form (Figure 14). It should be noted that we did not have access to the locations of crop cutting experiments, but only their values. This is why we opted for histogram comparisons between observed (crop cutting experiments) and modeled yield quantities. The frequency distributions of observed rice yields (blue bars) for several districts during the 2021 and 2022 Maha cropping season generally exhibit a unimodal shape, peaking between 3000 and 6000 kg/ha, depending on the district. Modeled yields (green bars) also show a unimodal distribution but generally have a larger range, particularly in the Maha 2021 season. The observed distribution appears somewhat narrower than the modeled distribution, indicating less variability in the observed yields. The slight discrepancy between observed and modeled quantities is even less pronounced in Maha 2022 across all districts examined here. To this end, the considerable overlap in the distributions, particularly with respect to their shape and the peak locations, lends further credibility to the significance of the regression model in estimating field-level yields across Sri Lanka under diverse geographic, climatic, and management conditions.

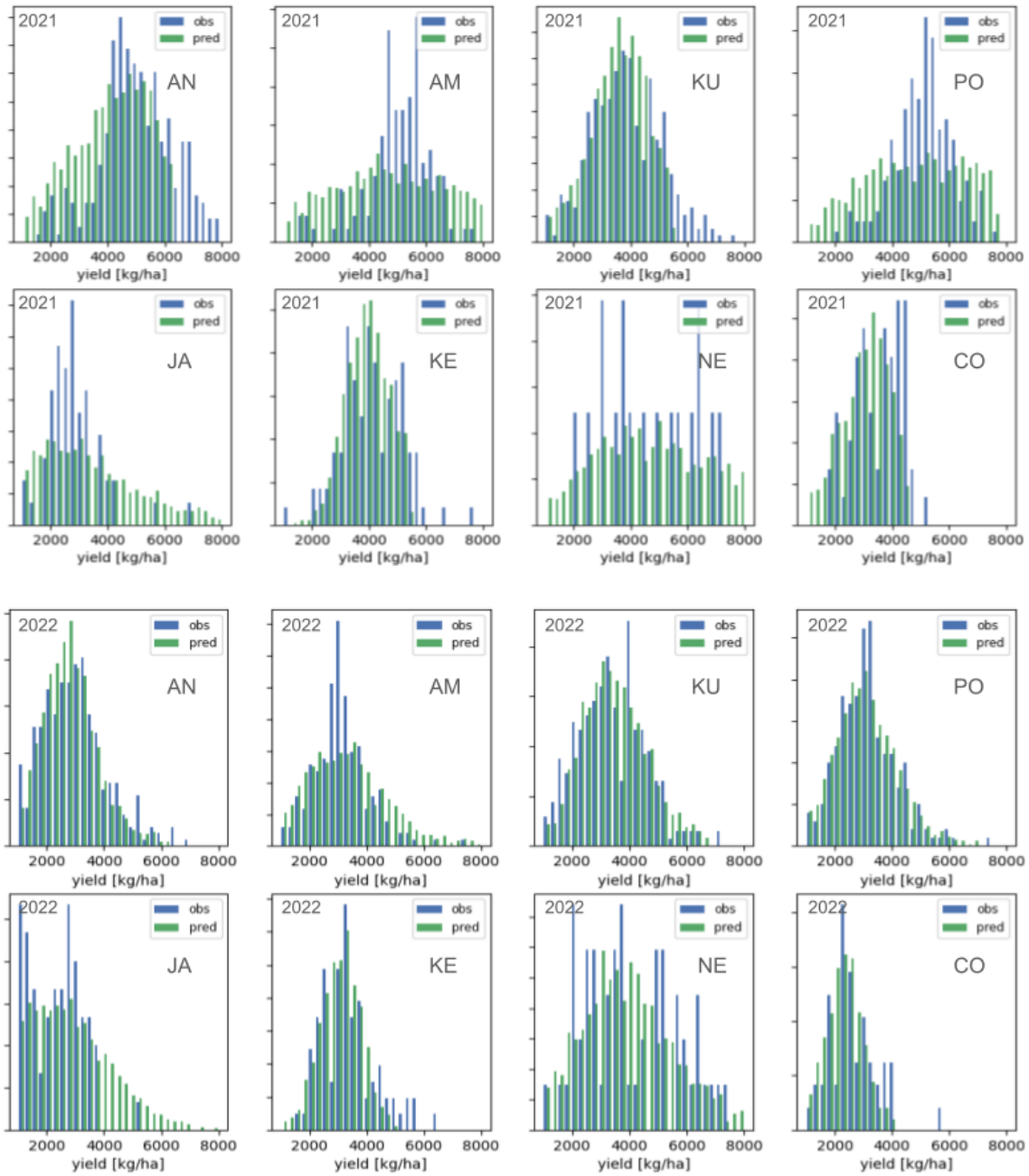


Figure 14. Comparison between observed (blue) and estimated (green) Maha rice yield histograms across eight districts known to contain the most (first and third rows) and the least (second and fourth rows) amount of rice in 2021 (top two rows) and 2022 (bottom two rows). The observed quantities are acquired from crop cutting experiments (DCS, 2023) [4]. The Y-axis in each plot reflects normalized counts. The estimated yield quantities are derived from a random sample in each district. The following districts are used in the comparison: AN: Anuradhapura; AM: Amara; KU: Kurunegala; PO: Polonnaruwa; JA: Jaffna; KE: Kegalle; NE: Nuwara Eliya; and CO: Colombo.

4. Discussion

In this paper, we presented a set of methods to map rice area and yield at field scales using optical remote sensing observations and limited training data in Sri Lanka. We employed an expert-based image classification algorithm on satellite observations that have been enhanced to isolate the rice signal based on the premise that paddy rice has growing conditions and phenology in Sri Lanka. To estimate crop yields, we used a machine learning regression model based on the random forest algorithm to combine

remotely sensed observations and environmental variables to produce pixel-level paddy rice yields across two decades (2000–2022). Our analysis reveals a remarkable degree of consistency between the remote sensing data and the government's statistics regarding rice cultivation, both in terms of area and yield. However, our work goes beyond and produces highly granular estimates of rice area and yield useful for understanding crop water use and methane emissions, as well as for quantifying the effectiveness of management and policy decisions.

This study highlights the power of optical remote sensing, even in locations like Sri Lanka with strong Monsoon seasons that have traditionally limited its application. This is mostly due to the availability of frequent and high-quality observations from Landsat and Sentinel-2 platforms that not only meet the demands of the phenology-driven rice algorithm presented here but also provide enough cloud-free observations to make it possible in the first place. Even with roughly 70% cloud cover during the main rice-growing season, there were sufficient observations to extract salient points in vegetation and moisture indices to identify rice fields and estimate their yields. This finding also suggests that historical (more than five years) rice area and yield estimates could be progressively less reliable than more recent (less than five years) estimates. Fusion of coarse-resolution and finer-scale observations may be one approach to remedy this issue [50,51]. Another obvious approach is to use SAR data to identify and map rice fields in such cloudy environments. In fact, there are many successful examples of this in the literature stretching back decades. However, consistent and repeated SAR observations are only available in the Sentinel-1 era (ca. 2016 to today) and hence their utility in rice mapping going back in time is somewhat limited. Our preliminary comparison of Sentinel-1 SAR observations against high-frequency optical observations from harmonized Sentinel-2 and Landsat models suggests an equal magnitude of errors made by each type, strengthening our proposal to rely on optical data, even in tropical regions with frequent cloud cover, such as Sri Lanka. Of course, the use of SAR data alone does not provide a viable method to estimate crop yields as the relationship between C-band backscatter and rice yields is not well-established [52,53].

While expert-based classification algorithms, as employed here, can be valuable for leveraging domain-specific knowledge in mapping rice fields in Sri Lanka, they may also be susceptible to several problems. A primary challenge lies in the inherent subjectivity and potential inconsistencies that arise when translating expert knowledge into definitive classification rules, which are employed in the form of thresholds in this study. For example, spectral signatures of rice can vary depending on growth stage, irrigation practices, and specific rice varieties, making it difficult for an expert to define universally applicable rules that capture this variability across Sri Lanka's diverse agro-ecological regions. Furthermore, rice paddies often exhibit spectral confusion with other land cover types, such as wetlands, and other flooded areas, especially during specific periods of the crop cycle or when using imagery with limited spectral resolution. This can lead to misclassification, particularly in complex, fragmented landscapes where smallholder farming dominates and rice fields are interspersed with other vegetation. The scalability of purely expert-driven approaches can also be problematic, as developing and refining rule sets for large areas or adapting them to changing conditions or new sensor data can be time-consuming and may not consistently capture the nuances of local cropping patterns and temporal dynamics without extensive, iterative adjustments, none of which were implemented here. While these issues are real and possibly present in the current study, the results also point to a successful and scalable implementation of the expert-based system employed here: errors of less than 20 percent while mapping rice fields with optical data in one of the most cloud-covered regions of the world is a remarkable achievement.

There are emerging sets of methods using deep learning to identify and map rice at field scales across many different environments [27,30]. While these methods present successful rice mapping examples, a closer examination suggests that (a) they are both applied in less cloud-prone areas such as northern China and (b) they rely on the availability of high-quality training samples to train and evaluate their models. In contrast, our strategy of relying on high-frequency optical observations and an expert-based and phenologically tuned algorithm that requires very little training data presents a great alternative. A survey of recent crop type mapping studies [26,54,55] suggests that methods relying on deep learning algorithms certainly push the envelope in our ability to identify and map crops in complicated landscapes, albeit with continued need for high-quality training data. To this end, perhaps one proposal would be to combine the approach we present here to automatically derive a large set of training data using phenological metrics associated with rice and use them in machine learning algorithms to produce high-quality maps. Future work could also integrate other climatic variables for improved yield forecasting, explore the transferability of our methodological framework to other regions, and investigate the socio-economic drivers behind observed changes in rice area.

The research presented here also provides a unique opportunity to assess the impact of the fertilizer import ban imposed by the government of Sri Lanka in 2021. We observe a notable drop (up to 40 percent in some areas) in rice yields during the 2021/2022 cropping season, indicating potential repercussions of the policy. Surprisingly, no significant changes in rice-cultivated areas are detected in the same period. This is most likely because most farmers had already committed to planting rice when the fertilizer ban went into effect. Moreover, rice cultivation in Sri Lanka relies heavily on irrigation systems (both large-scale and numerous small "tank" systems) designed for flooded paddy fields. Managing water for crops with different needs (e.g., less water overall or requiring better drainage rather than continuous flooding) becomes difficult as adapting existing irrigation schedules and on-farm water distribution for non-rice crops is a significant management hurdle. To this end, the results of this study underscore the potential for remote sensing as a powerful tool for assessing rice cultivation in Sri Lanka. The ability to accurately match government statistics with remote sensing data corroborates the technology's reliability and precision in monitoring agricultural activities. The remote sensing analysis also allows us to downscale existing government data to the scale of individual pixels that could be aggregated for economic analysis at any desired scale. The findings of this study also offer a valuable resource for policymakers in Sri Lanka. They provide a basis for evaluating the effectiveness (or lack thereof) of the fertilizer ban and its impact on rice production. While there is ample evidence in government records to show negative impacts on crop yields because of reduced fertilizer availability, this study produces additional spatial detail that may not be otherwise available. This information can guide future policy decisions regarding agriculture and food security.

5. Conclusions

This paper introduced a novel methodology for mapping rice area and yield at field scales in Sri Lanka, leveraging optical remote sensing data and minimal ground reference. We developed an expert-based image classification algorithm, specifically enhanced to isolate the unique phenological signal of paddy rice. For yield estimation, a random forest regression model was employed, integrating remotely sensed observations with environmental variables. The approach enabled us to generate two decades (2000–2022) of reliable, field-scale rice area and yield estimates in Sri Lanka with accuracies between 70% and 90% and yield estimations with less than 20% RMSE.

In conclusion, the use of remote sensing and an expert-based classification algorithm provided valuable insights into the spatial and temporal patterns of rice cultivation in Sri Lanka. Yield maps offered a window into spatial variation in production associated with soil, environmental conditions, and management practices, including the use of fertilizers across the country. Together, the methods described here allowed for the identification of geographic concentrations, shifts in cropping seasons (if any), and the adoption of farming practices as vehicles of change in rice-cultivated areas. The results revealed that remote sensing relying on optical data alone can successfully reproduce government statistics in terms of both area and yield in a tropical region with high cloud cover. However, the work presented here goes beyond and provides highly granular estimates that are useful for understanding crop water use and methane emissions, as well as for quantifying the effectiveness of management and policy decisions. As Sri Lanka continues to grapple with the effects of climate change and evolving agricultural demands, the data derived from remote sensing will remain indispensable for informed policy decisions and sustainable rice production practices in the country.

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Data Availability Statement: All data and code to reproduce rice and yield maps are publicly available at this repository: https://github.com/ozdogan15/srilanka#.

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Conflicts of Interest: The authors declare no conflicts of interest.

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